

Powershell - Variables & Logic

For actually automating with scripts (Like you would for Active Directory), you'd actually want to save *variables*; and use **If statements & For Loops**.

Variables are *saved, or called* via the \$ symbol. For example I create my "targetSize" variable, *then I print it out*.

```
$targetLength = 50MB  
Write-Host "Files larger than $targetLength"
```

- *I'm going to assume you already understand the underlying uses of variables*
-

Now, to **piggyback off** of the last PDF, we learned "|". This actually comes with a special variable, **\$_**

- What **\$_** does is *"act individually as every object the previous command outputs"*

Let's assume we have.

- (I know this looks daunting, but we're only here to understand the middle part)

```
Get-ChildItem -Path C:\Users\<yourUsername>\Downloads | Where-Object {  
$_ .Length -gt $targetLength } | Select-Object Name, Length
```

If you look at the "Where-Object" section, you'll see

- { \$_.Length -gt \$targetLength }

This takes EVERY object given from the **Get-ChildItem**, and checks if it is greater than (-gt) our target variable in length (50MB)

My Output →

Name	Length
----	-----
CursorUserSetup-x64-2.4.37.exe	178097128
Git-2.53.0-64-bit.exe	64573288

OK, how about we actually *do something with information* instead of just seeing it?

Move files if **they're more than 50MB to one folder**, and to another folder if they're *less than 50MB*.

```
Get-ChildItem -Path C:\Users\Aslam\Downloads -File -Recurse | ForEach-Object {  
    if ($_.Length -gt 50MB) {  
        Move-Item -Path $_.FullName -Destination "C:\Users\Aslam\Downloads\Big Folder"  
    } else {  
        Move-Item -Path $_.FullName -Destination "C:\Users\Aslam\Downloads\Smaller Files"  
    }  
}
```

First we grab every File (-File) inside our **downloads folder**, and in any folders inside downloads (-Recurse)

- We take **this first command**, add the "|" to *pass it into* "ForEach-Object" Which we need to do the IF statements,

Which is why the IF statement is INSIDE the ForEach-Object {}

Which brings us into the **If Statement**,

- *where we check the condition of a file being bigger than 50MB (\$_.Length -gt 50MB)*

The syntax for which is

```
if (condition) { run this  
} else { run this }
```

Then we use the **new command**, Move-Item

- It takes the **.FullName** (including the extension of the file **.pdf** or whatever)
 - Moves it into the **"Big Folder"** or **"Smaller Files"** accordingly
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So far we've just been covering syntax and the structure, next part we'll cover how to actually use common important commands with their arguments.